VALIDATION AND VERIFICATION

SCL PROPULSOR

MECHTRON 4TB6A

Group 11

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Revision History

Date	Version	Notes
February 12, 2025	1.0	Draft of V&V Document
March 23rd,2025	2.0	V & V Finalization

Background

The project seeks to create a robust testing framework for an adaptive cruise control algorithm, incorporating Software-in-the-Loop (SIL), Hardware-in-the-Loop (HIL), and a real-time visualizer. An RC (radio-controlled) vehicle serves as the testing platform, enabling the simulation and assessment of autonomous driving algorithms and safety features in a controlled, scalable environment.

Purpose

The purpose of this document is to outline the procedures taken to ensure the validation and verification of the requirements for the project. This includes testing and simulation of the hardware, software, and mechanical components.

Scope

The document will reference the system requirements that were outlined to ensure that all requirements are accounted for in validation and verification testing. The document includes hardware and software validation, the verification test plan, unit-testing and integration testing results.

Glossary

Table 1. Glossary.

Term	Definition
ACC	Adaptive Cruise Control.
Steady State	The state where a variable has not deviated more than a threshold (See Steady State Threshold)

Table 2. Monitored & Controlled Variables.

Term	Definition
Monitored Variables	
V_E	The velocity of the ego (host) vehicle in $\frac{cm}{s}$.
V_L	The longitudinal velocity of the lead vehicle in $\frac{cm}{s}$.
V_S	The target velocity of the ego vehicle in $\frac{cm}{s}$.
V_{Err}	The absolute difference between set speed and ego velocity; $V_{Err} = \lvert V_S - V_E \rvert$
t_{Err}	The absolute difference between the desired time gap and the actual time gap in seconds; $t_{Err} = \lvert t_{gap} - \frac{D_L}{V_E} \rvert$
a_x	Longitudinal acceleration ($\frac{cm}{s^2}$), the first derivative of ego velocity; $a_x = \frac{dV_E}{dt}$
J_x	Longitudinal jerk, the second derivative of ego velocity ($\frac{cm}{s^3}$); $J_x = \frac{d^2V_E}{dt^2}$
$W_1; W_2$	The ego velocity according to wheel speed sensor 1 and 2 respectively ($\frac{cm}{s}$)

$\boldsymbol{W}$	The average of W_1 and W_2 ($\frac{cm}{s}$); $W = \frac{W_1+W_2}{2}$
$\boldsymbol{T_1; T_2}$	The ego velocity according to treadmill speed sensor 1 and 2 respectively ($\frac{cm}{s}$)
$\boldsymbol{T}$	The average of T_1 and T_2 ($\frac{cm}{s}$); T= $\frac{T_1+T_2}{2}$
$\boldsymbol{D_L}$	The distance from the ego vehicle to the lead vehicle (cm). Distance is relative to the ego vehicle.
$\boldsymbol{t_{failure}}$	The latest timestamp when System failure occurred (seconds)
$\boldsymbol{t_{request_sim}}$	The latest timestamp when a request was sent to the Simulator (seconds)
Controlled Variables	
Act_Req	The actuation request from the microcontroller to the RC car. This is a unitless value that ranges from 0-255 that correlates to torque applied to wheels.
State	The operational state of the ACC. $State = ACTIVE$ when the $V_E \geq 20\frac{cm}{s}$, and $INACTIVE$ otherwise.

Table 3. Constant Variables.

Variable	Definition
$\boldsymbol{V_{Thres}}$	The speed above which ACC is considered active
$\boldsymbol{Act_{Min}}$	The minimum allowed value for Act_Req (aside from 0)
$\boldsymbol{Act_{Max}}$	The maximum allowed value for Act_Req
$\boldsymbol{t_{gap}}$	The ideal time gap

Dyn_{max}	The maximum difference between wheel speeds and treadmill speed.

Table 4. Calibratable Variables.

Variable	Definition
D_Max	The maximum distance that the simulator will consider a lead vehicle present.
SS_{Max}	The maximum amount of acceptable deviation in a velocity for it to be defined as steady state.
$V_{ErrMaxSS}$	The maximum allowable set speed error when in steady state
V_{ErrMax}	The maximum allowable set speed error when not in state
V_{ErrSM}	The maximum allowable difference between the set speed and detected speed
$t_{ErrMaxSS}$	The maximum allowable time gap error when in steady state
t_{ErrMax}	The maximum allowable time gap error when not in steady state
a_{max}	The maximum allowable longitudinal acceleration
J_{max}	The maximum allowable longitudinal jerk
Act_Rate_{Max}	The maximum allowable rate of change of actuation request

Hardware Validation

Hardware validation for this project's requirements consisted of the ability for the mechanical and electrical components to accurately and appropriately respond in real-time for hardware-in-loop and software-in-loop environments. Requirements were created and followed through on everything that has been integrated such as appropriate serial communication between the on-board micro-controllers, integration of a safety monitor to the remote-controlled vehicle, and hall-effect sensors that measure wheel revolutions.

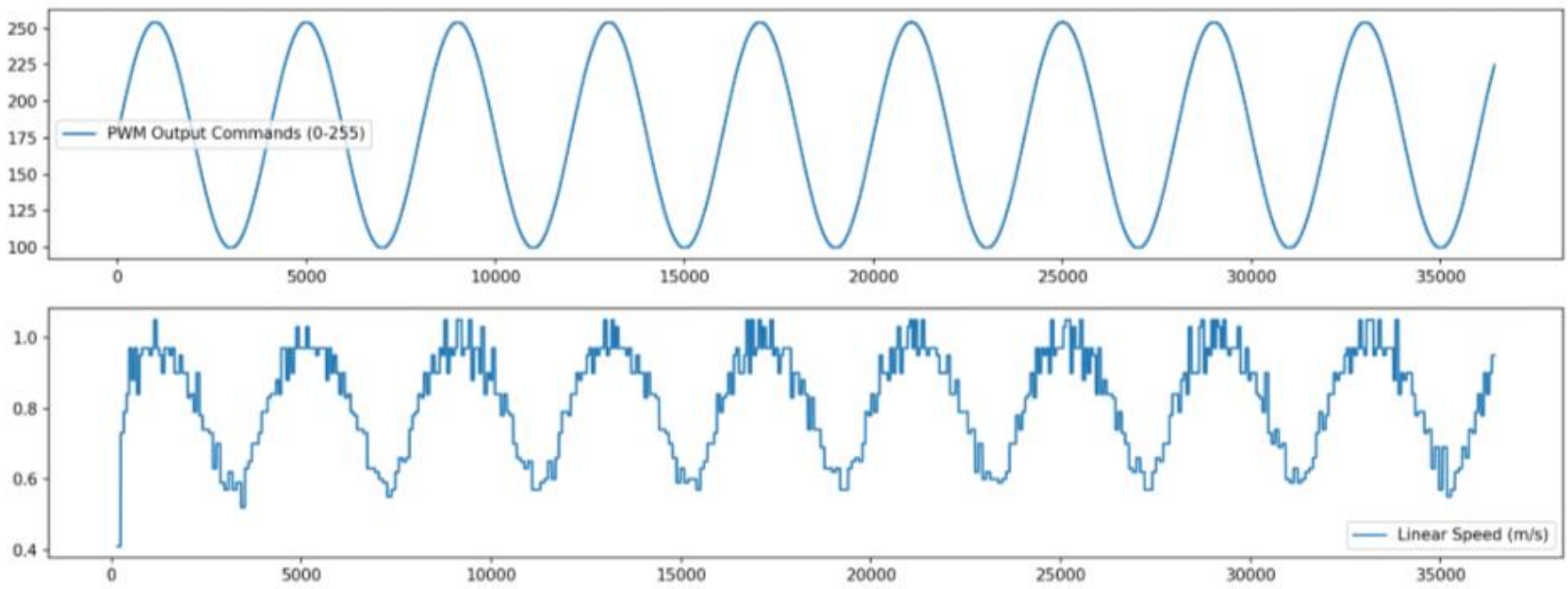


Figure 1. Validation of hall-effect wheel speed sensor readings.

The hardware requirements for the mechanical and electrical systems provided a closed-loop control that can relay real-time information back to the software in the loop environment. Moving forward, the requirements for the SimDyno will have to be checked upon fabrication to complete the hardware validation.

Software Validation

Software validation of this project consists of being able to run a robust simulation, the correctness of the ACC algorithm, and the correctness of the safety monitor algorithms. We will not be proving the correctness of any algorithms in this project due to pragmatism, though that it is ideal. Instead, we will validate these systems will behave accordingly via system and integration tests, as the software of these systems continuously requires input and outputs from each other as well as integration with the electronics.

Noting that the data flow between the software systems is fairly coupled, it would be most feasible to determine the validity of the software by testing with various types of data, as is done in our testing plan.

Furthermore, an integral part of the success of this project is the feasible use of it in McMaster EcoCar. As this project entirely consists of students working in EcoCar, we will be able to validate the entire system and evaluate its usability by using the system in its intended environment.

Hardware Testing Plan

The hardware testing plan will consist of numerous isolated tests on the electrical and mechanical components to ensure appropriate system behaviour. Testing consists of a combination of qualitative and quantitative measurements to ensure proper validation and verification of the project's requirements. This is due in part to the complexity and inaccessibility of proper testing facilities. Tests will be considered to pass if the system exhibits expected results. Unfortunately, due to component backlog, components that are to be used for the construction of the SimDyno were unable to be purchased and delivered. As a result, tests requiring the completion of the SimDyno will be incomplete and completed later.

Software Testing Plan

Our testing process will be split into two phases: system tests and unit tests. Many of the requirements associated with Adaptive Cruise Control are the result of multiple different modules working together. For that purpose, system tests were designed to verify the requirements. These involve generating a specific set of inputs, called test sequences, which simulate the behaviour of the lead vehicle and the set speed.

Five test sequences each were designed for the lead vehicle behaviour, and set speed, for a total of 25 unique tests. Each combination of these sequences will be run, and the results will be compared against the expected results based on the relevant requirements. Additionally, the last test sequence for each set is random, so multiple runs of those will be done to highlight any unique bugs.

Many of the other modules in our project are better tested individually, using unit tests. For each of these, a specific test was designed for each requirement.

System Tests

Test Sequences

Lead Vehicle Test Sequences

These test sequences will define the behaviour of the lead vehicle. All will take the form of a list of Lead Velocity values and their associated time steps. For all test sequences but the first one, the Lead Distance will start at 1m then be calculated at each time step based on the difference between the Lead and Ego Velocities.

LV1. No Lead Vehicle – Lead Velocity and Lead Distance will be set as arbitrarily high values such that the lead vehicle is never relevant. This will be used to test the Cruise Control (non-adaptive) functionality.
LV2. Standard Highway – The Lead Velocity will be taken from the HWFET drive cycle, a standardized drive cycle meant to simulate typical highway driving. This will be scaled down for the smaller speeds of the RC car.
LV3. Standard City – Similar to the Standard Highway Test Sequence, this will use the UDDS drive cycle, which emulates city driving.
LV4. Stepwise Velocity – The Lead Velocity will take on values of 0.4 m/s, 0.55 m/s, 1 m/s, 0.55 m/s, 0.4 m/s for 30 seconds each, in that order. This sequence will test step responses both upwards and downwards.
LV5. Erratic – The simulator will randomly generate a value between 0.02 – 0.08 m/s for every 10 seconds of run time. The Lead Velocity will be a connection of these values using a third order polynomial. This sequence will test the most extreme behaviour of the lead vehicle. Because this test sequence is non-deterministic, multiple instances of this test will need to be run to increase the confidence of its results.

Set Speed Test Sequences

These test sequences will define the value of the Set Speed. Each will be a piecewise constant function.
SS1. Constant Average – The Set Speed will be 0.55 m/s for the entirety of the test.
SS2. Constant Minimum – The Set Speed will be 0.4 m/s for the entirety of the test.
SS3. Constant Maximum – The Set Speed will be 1 m/s for the entirety of the test.
SS4. Stepwise – The Set Speed will take on values of 1 m/s, 0.55 m/s, 0.4 m/s, 0.55 m/s, 1 m/s for 30 seconds each, in that order. This sequence will test step responses both upwards and downwards.

SS5. Random – The set speed will take on any value within 0.2 – 0.8 m/s, changing step-wise at random intervals, ranging from 5 to 30 seconds. Again, because this test sequence is non-deterministic, multiple instances of this test will need to be run to increase the confidence of its results.

The graphical results of all of these SIL tests are shown in Appendix A. Tests marked as passed have either explicitly passed the requirement, or the outputs were validated and a simulation-specific behavior (from unrepresentative plant model or abnormally high transients) where the test failed according to the requirement, but the high-level behavior is acceptable.

System Test Traceability Matrix

LV1 – No Lead Vehicle

Table 5. LV1 No Lead Vehicle Traceability Matrix.

Test ID	Expected Outputs	Associated Requirement(s)	Result	Test Status
ACC_1.1-LV1-SS1	$While\ (V_E < V_{Thres})$, $Act_Req = Act_{Min}$	ACC_1.1	*Passed*	*Complete*
ACC_1.2-LV1-SS1	$While\ (V_E < V_{Thres})$, $State = INACTIVE$	ACC_1.2	*Passed*	*Complete*
ACC_1.3-LV1-SS1	$While\ (V_E \geq V_{Thres})$, $State = ACTIVE$	ACC_1.3	*Passed*	*Complete*
ACC_2.1-LV1-SS1	$V_{Err} \leq V_{ErrMaxSS}$	ACC_2.1	*Passed*	*Complete*
ACC_3.1-LV1-SS1	$While(State = Active)$, $\lvert a_x \rvert \leq a_{Max}$	ACC_3.1	*Passed*	*Complete*
ACC_3.2-LV1-SS1	$While(State = Active)$, $\lvert J_x \rvert \leq J_{Max}$	ACC_3.2	*Passed*	*Complete*
ACC_5.1-LV1-SS1	$While(State = Active)$, $100 \leq Act_\mathrm{Req} \leq 255$	ACC_5.1	*Passed*	*Complete*

SM_3.1-LV1-SS1	$While(State = Active)$, $\lvert Act_Req(t) - Act_Req(t-1)\rvert \leq Act_Rate_{Max}$	SM_3.1 (and other similar)	*Passed*	*Complete*
ACC_2.1-LV1-SS2	$V_{Err} \leq V_{ErrMaxSS}$	ACC_2.1	*Passed*	*Complete*
ACC_3.1-LV1-SS2	$While(State = Active)$, $\lvert a_x\rvert \leq a_{Max}$	ACC_3.1	*Passed*	*Complete*
ACC_3.2-LV1-SS2	$While(State = Active)$, $\lvert J_x\rvert \leq J_{Max}$	ACC_3.2	*Passed*	*Complete*
ACC_5.1-LV1-SS2	$While(State = Active)$, $100 \leq Act_\text{Req} \leq 255$	ACC_5.1	*Passed*	*Complete*
SM_3.1-LV1-SS2	$While(State = Active)$, $\lvert Act_Req(t) - Act_Req(t-1)\rvert \leq Act_Rate_{Max}$	SM_3.1 (and other similar)	*Passed*	*Complete*
ACC_2.1-LV1-SS3	$V_{Err} \leq V_{ErrMaxSS}$	ACC_2.1	*Passed*	*Complete*
ACC_3.1-LV1-SS3	$While(State = Active)$, $\lvert a_x\rvert \leq a_{Max}$	ACC_3.1	*Passed*	*Complete*
ACC_3.2-LV1-SS3	$While(State = Active)$, $\lvert J_x\rvert \leq J_{Max}$	ACC_3.2	*Passed*	*Complete*
ACC_5.1-LV1-SS3	$While(State = Active)$, $100 \leq Act_\text{Req} \leq 255$	ACC_5.1	*Passed*	*Complete*
SM_3.1-LV1-SS3	$While(State = Active)$, $\lvert Act_Req(t) - Act_Req(t-1)\rvert \leq Act_Rate_{Max}$	SM_3.1 (and other similar)	*Passed*	*Complete*
ACC_2.1-LV1-SS4	$While\ (\lvert V_S(t) - V_S(t-1)\rvert \leq SS_{Max})$, $V_{Err} \leq V_{ErrMaxSS}$	ACC_2.1	*Passed*	*Complete*
ACC_2.2-LV1-SS4	$While\ (\lvert V_S(t) - V_S(t-1)\rvert > SS_{Max})$, $V_{Err} \leq V_{ErrMax}$	ACC_2.2	*Passed*	*Complete*
ACC_3.1-LV1-SS4	$While(State = Active)$, $\lvert a_x\rvert \leq a_{Max}$	ACC_3.1	*Passed*	*Complete*

ACC_3.2-LV1-SS4	$While(State = Active)$, $\lvert J_x \rvert \leq J_{Max}$	ACC_3.2	*Passed*	*Complete*
ACC_5.1-LV1-SS4	$While(State = Active)$, $100 \leq Act_\text{Req} \leq 255$	ACC_5.1	*Passed*	*Complete*
SM_3.1-LV1-SS4	$While(State = Active)$, $\lvert Act_Req(t) - Act_Req(t-1) \rvert \leq Act_Rate_{Max}$	SM_3.1 (and other similar)	*Passed*	*Complete*
ACC_2.1-LV1-SS5	$While\ (\lvert V_S(t) - V_S(t-1) \rvert \leq SS_{Max})$, $V_{Err} \leq V_{ErrMaxSS}$	ACC_2.1	*Passed*	*Complete*
ACC_2.2-LV1-SS5	$While\ (\lvert V_S(t) - V_S(t-1) \rvert > SS_{Max})$, $V_{Err} \leq V_{ErrMax}$	ACC_2.2	*Passed*	*Complete*
ACC_3.1-LV1-SS5	$While(State = Active)$, $\lvert a_x \rvert \leq a_{Max}$	ACC_3.1	*Passed*	*Complete*
ACC_3.2-LV1-SS5	$While(State = Active)$, $\lvert J_x \rvert \leq J_{Max}$	ACC_3.2	*Passed*	*Complete*
ACC_5.1-LV1-SS5	$While(State = Active)$, $100 \leq Act_\text{Req} \leq 255$	ACC_5.1	*Passed*	*Complete*
SM_3.1-LV1-SS5	$While(State = Active)$, $\lvert Act_Req(t) - Act_Req(t-1) \rvert \leq Act_Rate_{Max}$	SM_3.1 (and other similar)	*Passed*	*Complete*

LV2 – Standard Highway

Table 6. LV2 Standard Highway Traceability Matrix.

Test ID	**Expected Outputs**	**Associated Requirement(s)**	**Result**	**Test Status**
ACC_2.1-LV2-SS1	$While\ ((D_L > D_{Max}\ or\ V_S > V_L)\ and\ \lvert V_S(t) - V_S(t-1) \rvert \leq SS_{Max})$, $V_{Err} \leq V_{ErrMaxSS}$	ACC_2.1	*Passed*	*Complete*

ACC_2.2-LV2-SS1	$While\ ((D_L > D_{Max}\ or\ V_S > V_L)\ and\ \lvert V_S(t) - V_S(t-1)\rvert > SS_{Max}),\ V_{Err} \leq V_{ErrMax}$	ACC_2.2	*Passed*	*Complete*
ACC_2.3-LV2-SS1	$While\ (D_L \leq D_{Max}\ and\ V_S \leq V_L\ and\ \lvert V_S(t) - V_S(t-1)\rvert \leq SS_{Max}),\ V_{Err} \leq V_{ErrMaxSS}$	ACC_2.3	*Passed*	*Complete*
ACC_2.4-LV2-SS1	$While\ (D_L \leq D_{Max}\ and\ V_S \leq V_L\ and\ \lvert V_S(t) - V_S(t-1)\rvert > SS_{Max}),\ V_{Err} \leq V_{ErrMax}$	ACC_2.4	*Passed*	*Complete*
ACC_3.1-LV2-SS1	$While(State = Active),\ \lvert a_x\rvert \leq a_{Max}$	ACC_3.1	*Passed*	*Complete*
ACC_3.2-LV2-SS1	$While(State = Active),\ \lvert J_x\rvert \leq J_{Max}$	ACC_3.2	*Passed*	*Complete*
ACC_5.1-LV2-SS1	$While(State = Active),\ 100 \leq Act_\mathrm{Req} \leq 255$	ACC_5.1	*Passed*	*Complete*
SM_3.1-LV2-SS1	$While(State = Active),\ \lvert Act_Req(t) - Act_Req(t-1)\rvert \leq Act_Rate_{Max}$	SM_3.1 (and other similar)	*Passed*	*Complete*
ACC_2.1-LV2-SS2	$While\ ((D_L > D_{Max}\ or\ V_S > V_L)\ and\ \lvert V_S(t) - V_S(t-1)\rvert \leq SS_{Max}),\ V_{Err} \leq V_{ErrMaxSS}$	ACC_2.1	*Passed*	*Complete*
ACC_2.2-LV2-SS2	$While\ ((D_L > D_{Max}\ or\ V_S > V_L)\ and\ \lvert V_S(t) - V_S(t-1)\rvert > SS_{Max}),\ V_{Err} \leq V_{ErrMax}$	ACC_2.2	*Passed*	*Complete*
ACC_2.3-LV2-SS2	$While\ (D_L \leq D_{Max}\ and\ V_S \leq V_L\ and\ \lvert V_S(t) - V_S(t-1)\rvert \leq SS_{Max}),\ V_{Err} \leq V_{ErrMaxSS}$	ACC_2.3	*Passed*	*Complete*
ACC_2.4-LV2-SS2	$While\ (D_L \leq D_{Max}\ and\ V_S \leq V_L\ and\ \lvert V_S(t) - V_S(t-1)\rvert > SS_{Max}),\ V_{Err} \leq V_{ErrMax}$	ACC_2.4	*Passed*	*Complete*
ACC_3.1-LV2-SS2	$While(State = Active),$	ACC_3.1	*Passed*	*Complete*

	$\|a_x\| \leq a_{Max}$			
ACC_3.2-LV2-SS2	$While(State = Active)$, $\|J_x\| \leq J_{Max}$	ACC_3.2	*Passed*	*Complete*
ACC_5.1-LV2-SS2	$While(State = Active)$, $100 \leq Act_\text{Req} \leq 255$	ACC_5.1	*Passed*	*Complete*
SM_3.1-LV2-SS2	$While(State = Active)$, $\|Act_Req(t) - Act_Req(t-1)\| \leq Act_Rate_{Max}$	SM_3.1 (and other similar)	*Passed*	*Complete*
ACC_2.1-LV2-SS3	$While\ ((D_L > D_{Max}\ or\ V_S > V_L)\ and\ \|V_S(t) - V_S(t-1)\| \leq SS_{Max})$, $V_{Err} \leq V_{ErrMaxSS}$	ACC_2.1	*Passed*	*Complete*
ACC_2.2-LV2-SS3	$While\ ((D_L > D_{Max}\ or\ V_S > V_L)\ and\ \|V_S(t) - V_S(t-1)\| > SS_{Max})$, $V_{Err} \leq V_{ErrMax}$	ACC_2.2	*Passed*	*Complete*
ACC_2.3-LV2-SS3	$While\ (D_L \leq D_{Max}\ and\ V_S \leq V_L\ and\ \|V_S(t) - V_S(t-1)\| \leq SS_{Max})$, $V_{Err} \leq V_{ErrMaxSS}$	ACC_2.3	*Passed*	*Complete*
ACC_2.4-LV2-SS3	$While\ (D_L \leq D_{Max}\ and\ V_S \leq V_L\ and\ \|V_S(t) - V_S(t-1)\| > SS_{Max})$, $V_{Err} \leq V_{ErrMax}$	ACC_2.4	*Passed*	*Complete*
ACC_3.1-LV2-SS3	$While(State = Active)$, $\|a_x\| \leq a_{Max}$	ACC_3.1	*Passed*	*Complete*
ACC_3.2-LV2-SS3	$While(State = Active)$, $\|J_x\| \leq J_{Max}$	ACC_3.2	*Passed*	*Complete*
ACC_5.1-LV2-SS3	$While(State = Active)$, $100 \leq Act_\text{Req} \leq 255$	ACC_5.1	*Passed*	*Complete*
SM_3.1-LV2-SS3	$While(State = Active)$, $\|Act_Req(t) - Act_Req(t-1)\| \leq Act_Rate_{Max}$	SM_3.1 (and other similar)	*Passed*	*Complete*
ACC_2.1-LV2-SS4	$While\ ((D_L > D_{Max}\ or\ V_S > V_L)\ and\ \|V_S(t) - V_S(t-1)\| \leq SS_{Max})$,	ACC_2.1	*Passed*	*Complete*

	$V_{Err} \leq V_{ErrMaxSS}$			
ACC_2.2-LV2-SS4	$While\ ((D_L > D_{Max}\ or\ V_S > V_L)\ and\ \lvert V_S(t) - V_S(t-1)\rvert > SS_{Max}),\ V_{Err} \leq V_{ErrMax}$	ACC_2.2	*Passed*	*Complete*
ACC_2.3-LV2-SS4	$While\ (D_L \leq D_{Max}\ and\ V_S \leq V_L\ and\ \lvert V_S(t) - V_S(t-1)\rvert \leq SS_{Max}),\ V_{Err} \leq V_{ErrMaxSS}$	ACC_2.3	*Passed*	*Complete*
ACC_2.4-LV2-SS4	$While\ (D_L \leq D_{Max}\ and\ V_S \leq V_L\ and\ \lvert V_S(t) - V_S(t-1)\rvert > SS_{Max}),\ V_{Err} \leq V_{ErrMax}$	ACC_2.4	*Passed*	*Complete*
ACC_3.1-LV2-SS4	$While(State = Active),\ \lvert a_x\rvert \leq a_{Max}$	ACC_3.1	*Passed*	*Complete*
ACC_3.2-LV2-SS4	$While(State = Active),\ \lvert J_x\rvert \leq J_{Max}$	ACC_3.2	*Passed*	*Complete*
ACC_5.1-LV2-SS4	$While(State = Active),\ 100 \leq Act_\mathrm{Req} \leq 255$	ACC_5.1	*Passed*	*Complete*
SM_3.1-LV2-SS4	$While(State = Active),\ \lvert Act_Req(t) - Act_Req(t-1)\rvert \leq Act_Rate_{Max}$	SM_3.1 (and other similar)	*Passed*	*Complete*
ACC_2.1-LV2-SS1	$While\ ((D_L > D_{Max}\ or\ V_S > V_L)\ and\ \lvert V_S(t) - V_S(t-1)\rvert \leq SS_{Max}),\ V_{Err} \leq V_{ErrMaxSS}$	ACC_2.1	*Passed*	*Complete*
ACC_2.2-LV2-SS1	$While\ ((D_L > D_{Max}\ or\ V_S > V_L)\ and\ \lvert V_S(t) - V_S(t-1)\rvert > SS_{Max}),\ V_{Err} \leq V_{ErrMax}$	ACC_2.2	*Passed*	*Complete*
ACC_2.3-LV2-SS1	$While\ (D_L \leq D_{Max}\ and\ V_S \leq V_L\ and\ \lvert V_S(t) - V_S(t-1)\rvert \leq SS_{Max}),\ V_{Err} \leq V_{ErrMaxSS}$	ACC_2.3	*Passed*	*Complete*
ACC_2.4-LV2-SS1	$While\ (D_L \leq D_{Max}\ and\ V_S \leq V_L\ and\ \lvert V_S(t) - V_S(t-1)\rvert > SS_{Max}),\ V_{Err} \leq V_{ErrMax}$	ACC_2.4	*Passed*	*Complete*

ACC_3.1-LV2-SS5	$While(State = Active)$, $\lvert a_x \rvert \leq a_{Max}$	ACC_3.1	*Passed*	*Complete*
ACC_3.2-LV2-SS5	$While(State = Active)$, $\lvert J_x \rvert \leq J_{Max}$	ACC_3.2	*Passed*	*Complete*
ACC_5.1-LV2-SS5	$While(State = Active)$, $100 \leq Act_\mathrm{Req} \leq 255$	ACC_5.1	*Passed*	*Complete*
SM_3.1-LV2-SS5	$While(State = Active)$, $\lvert Act_Req(t) - Act_Req(t-1) \rvert \leq Act_Rate_{Max}$	SM_3.1 (and other similar)	*Passed*	*Complete*

LV3 – Standard City

LV3 – LV5 have the same set of tests as LV2 so everything aside from the test results are duplicates of that table.

Table 7. LV3 Standard City Traceability Matrix.

Test ID	**Expected Outputs**	**Associated Requirement(s)**	**Result**	**Test Status**
ACC_2.1-LV3-SS1	$While\ ((D_L > D_{Max}\ or\ V_S > V_L)\ and\ \lvert V_S(t) - V_S(t-1) \rvert \leq SS_{Max})$, $V_{Err} \leq V_{ErrMaxSS}$	ACC_2.1	*Passed*	*Complete*
ACC_2.2-LV3-SS1	$While\ ((D_L > D_{Max}\ or\ V_S > V_L)\ and\ \lvert V_S(t) - V_S(t-1) \rvert > SS_{Max})$, $V_{Err} \leq V_{ErrMax}$	ACC_2.2	*Passed*	*Complete*
ACC_2.3-LV3-SS1	$While\ (D_L \leq D_{Max}\ and\ V_S \leq V_L\ and\ \lvert V_S(t) - V_S(t-1) \rvert \leq SS_{Max})$, $V_{Err} \leq V_{ErrMaxSS}$	ACC_2.3	*Passed*	*Complete*
ACC_2.4-LV3-SS1	$While\ (D_L \leq D_{Max}\ and\ V_S \leq V_L\ and\ \lvert V_S(t) - V_S(t-1) \rvert > SS_{Max})$, $V_{Err} \leq V_{ErrMax}$	ACC_2.4	*Passed*	*Complete*
ACC_3.1-LV3-SS1	$While(State = Active)$,	ACC_3.1	*Passed*	*Complete*

	$\|a_x\| \leq a_{Max}$			
ACC_3.2-LV3-SS1	$While(State = Active)$, $\|J_x\| \leq J_{Max}$	ACC_3.2	*Passed*	*Complete*
ACC_5.1-LV3-SS1	$While(State = Active)$, $100 \leq Act_\text{Req} \leq 255$	ACC_5.1	*Passed*	*Complete*
SM_3.1-LV3-SS1	$While(State = Active)$, $\|Act_Req(t) - Act_Req(t-1)\| \leq Act_Rate_{Max}$	SM_3.1 (and other similar)	*Passed*	*Complete*
ACC_2.1-LV3-SS2	$While\ ((D_L > D_{Max}\ or\ V_S > V_L)\ and\ \|V_S(t) - V_S(t-1)\| \leq SS_{Max})$, $V_{Err} \leq V_{ErrMaxSS}$	ACC_2.1	*Passed*	*Complete*
ACC_2.2-LV3-SS2	$While\ ((D_L > D_{Max}\ or\ V_S > V_L)\ and\ \|V_S(t) - V_S(t-1)\| > SS_{Max})$, $V_{Err} \leq V_{ErrMax}$	ACC_2.2	*Passed*	*Complete*
ACC_2.3-LV3-SS2	$While\ (D_L \leq D_{Max}\ and\ V_S \leq V_L\ and\ \|V_S(t) - V_S(t-1)\| \leq SS_{Max})$, $V_{Err} \leq V_{ErrMaxSS}$	ACC_2.3	*Passed*	*Complete*
ACC_2.4-LV3-SS2	$While\ (D_L \leq D_{Max}\ and\ V_S \leq V_L\ and\ \|V_S(t) - V_S(t-1)\| > SS_{Max})$, $V_{Err} \leq V_{ErrMax}$	ACC_2.4	*Passed*	*Complete*
ACC_3.1-LV3-SS2	$While(State = Active)$, $\|a_x\| \leq a_{Max}$	ACC_3.1	*Passed*	*Complete*
ACC_3.2-LV3-SS2	$While(State = Active)$, $\|J_x\| \leq J_{Max}$	ACC_3.2	*Passed*	*Complete*
ACC_5.1-LV3-SS2	$While(State = Active)$, $100 \leq Act_\text{Req} \leq 255$	ACC_5.1	*Passed*	*Complete*
SM_3.1-LV3-SS2	$While(State = Active)$, $\|Act_Req(t) - Act_Req(t-1)\| \leq Act_Rate_{Max}$	SM_3.1 (and other similar)	*Passed*	*Complete*
ACC_2.1-LV3-SS3	$While\ ((D_L > D_{Max}\ or\ V_S > V_L)\ and\ \|V_S(t) - V_S(t-1)\| \leq SS_{Max})$,	ACC_2.1	*Passed*	*Complete*

	$V_{Err} \leq V_{ErrMaxSS}$			
ACC_2.2-LV3-SS3	$While\ ((D_L > D_{Max}\ or\ V_S > V_L)\ and\ \lvert V_S(t) - V_S(t-1)\rvert > SS_{Max}),\ V_{Err} \leq V_{ErrMax}$	ACC_2.2	*Passed*	*Complete*
ACC_2.3-LV3-SS3	$While\ (D_L \leq D_{Max}\ and\ V_S \leq V_L\ and\ \lvert V_S(t) - V_S(t-1)\rvert \leq SS_{Max}),\ V_{Err} \leq V_{ErrMaxSS}$	ACC_2.3	*Passed*	*Complete*
ACC_2.4-LV3-SS3	$While\ (D_L \leq D_{Max}\ and\ V_S \leq V_L\ and\ \lvert V_S(t) - V_S(t-1)\rvert > SS_{Max}),\ V_{Err} \leq V_{ErrMax}$	ACC_2.4	*Passed*	*Complete*
ACC_3.1-LV3-SS3	$While(State = Active),\ \lvert a_x\rvert \leq a_{Max}$	ACC_3.1	*Passed*	*Complete*
ACC_3.2-LV3-SS3	$While(State = Active),\ \lvert J_x\rvert \leq J_{Max}$	ACC_3.2	*Passed*	*Complete*
ACC_5.1-LV3-SS3	$While(State = Active),\ 100 \leq Act_\mathrm{Req} \leq 255$	ACC_5.1	*Passed*	*Complete*
SM_3.1-LV3-SS3	$While(State = Active),\ \lvert Act_Req(t) - Act_Req(t-1)\rvert \leq Act_Rate_{Max}$	SM_3.1 (and other similar)	*Passed*	*Complete*
ACC_2.1-LV3-SS4	$While\ ((D_L > D_{Max}\ or\ V_S > V_L)\ and\ \lvert V_S(t) - V_S(t-1)\rvert \leq SS_{Max}),\ V_{Err} \leq V_{ErrMaxSS}$	ACC_2.1	*Passed*	*Complete*
ACC_2.2-LV3-SS4	$While\ ((D_L > D_{Max}\ or\ V_S > V_L)\ and\ \lvert V_S(t) - V_S(t-1)\rvert > SS_{Max}),\ V_{Err} \leq V_{ErrMax}$	ACC_2.2	*Passed*	*Complete*
ACC_2.3-LV3-SS4	$While\ (D_L \leq D_{Max}\ and\ V_S \leq V_L\ and\ \lvert V_S(t) - V_S(t-1)\rvert \leq SS_{Max}),\ V_{Err} \leq V_{ErrMaxSS}$	ACC_2.3	*Passed*	*Complete*
ACC_2.4-LV3-SS4	$While\ (D_L \leq D_{Max}\ and\ V_S \leq V_L\ and\ \lvert V_S(t) - V_S(t-1)\rvert > SS_{Max}),\ V_{Err} \leq V_{ErrMax}$	ACC_2.4	*Passed*	*Complete*

ACC_3.1-LV3-SS4	$While(State = Active)$, $\lvert a_x \rvert \leq a_{Max}$	ACC_3.1	*Passed*	*Complete*
ACC_3.2-LV3-SS4	$While(State = Active)$, $\lvert J_x \rvert \leq J_{Max}$	ACC_3.2	*Passed*	*Complete*
ACC_5.1-LV3-SS4	$While(State = Active)$, $100 \leq Act_\text{Req} \leq 255$	ACC_5.1	*Passed*	*Complete*
SM_3.1-LV3-SS4	$While(State = Active)$, $\lvert Act_Req(t) - Act_Req(t-1) \rvert \leq Act_Rate_{Max}$	SM_3.1 (and other similar)	*Passed*	*Complete*
ACC_2.1-LV3-SS1	$While\ ((D_L > D_{Max}\ or\ V_S > V_L)\ and\ \lvert V_S(t) - V_S(t-1) \rvert \leq SS_{Max})$, $V_{Err} \leq V_{ErrMaxSS}$	ACC_2.1	*Passed*	*Complete*
ACC_2.2-LV3-SS1	$While\ ((D_L > D_{Max}\ or\ V_S > V_L)\ and\ \lvert V_S(t) - V_S(t-1) \rvert > SS_{Max})$, $V_{Err} \leq V_{ErrMax}$	ACC_2.2	*Passed*	*Complete*
ACC_2.3-LV3-SS1	$While\ (D_L \leq D_{Max}\ and\ V_S \leq V_L\ and\ \lvert V_S(t) - V_S(t-1) \rvert \leq SS_{Max})$, $V_{Err} \leq V_{ErrMaxSS}$	ACC_2.3	*Passed*	*Complete*
ACC_2.4-LV3-SS1	$While\ (D_L \leq D_{Max}\ and\ V_S \leq V_L\ and\ \lvert V_S(t) - V_S(t-1) \rvert > SS_{Max})$, $V_{Err} \leq V_{ErrMax}$	ACC_2.4	*Passed*	*Complete*
ACC_3.1-LV3-SS5	$While(State = Active)$, $\lvert a_x \rvert \leq a_{Max}$	ACC_3.1	*Passed*	*Complete*
ACC_3.2-LV3-SS5	$While(State = Active)$, $\lvert J_x \rvert \leq J_{Max}$	ACC_3.2	*Passed*	*Complete*
ACC_5.1-LV3-SS5	$While(State = Active)$, $100 \leq Act_\text{Req} \leq 255$	ACC_5.1	*Passed*	*Complete*
SM_3.1-LV3-SS5	$While(State = Active)$, $\lvert Act_Req(t) - Act_Req(t-1) \rvert \leq Act_Rate_{Max}$	SM_3.1 (and other similar)	*Passed*	*Complete*

LV4 – Stepwise Velocity

Table 8. LV4 Stepwise Velocity Traceability Matrix.

Test ID	Expected Outputs	Associated Requirement(s)	Result	Test Status
ACC_2.1-LV4-SS1	$While\ ((D_L > D_{Max}\ or\ V_S > V_L)\ and\ \lvert V_S(t) - V_S(t-1)\rvert \leq SS_{Max}),\ V_{Err} \leq V_{ErrMaxSS}$	ACC_2.1	*Passed*	*Complete*
ACC_2.2-LV4-SS1	$While\ ((D_L > D_{Max}\ or\ V_S > V_L)\ and\ \lvert V_S(t) - V_S(t-1)\rvert > SS_{Max}),\ V_{Err} \leq V_{ErrMax}$	ACC_2.2	*Passed*	*Complete*
ACC_2.3-LV4-SS1	$While\ (D_L \leq D_{Max}\ and\ V_S \leq V_L\ and\ \lvert V_S(t) - V_S(t-1)\rvert \leq SS_{Max}),\ V_{Err} \leq V_{ErrMaxSS}$	ACC_2.3	*Passed*	*Complete*
ACC_2.4-LV4-SS1	$While\ (D_L \leq D_{Max}\ and\ V_S \leq V_L\ and\ \lvert V_S(t) - V_S(t-1)\rvert > SS_{Max}),\ V_{Err} \leq V_{ErrMax}$	ACC_2.4	*Passed*	*Complete*
ACC_3.1-LV4-SS1	$While(State = Active),\ \lvert a_x\rvert \leq a_{Max}$	ACC_3.1	*Passed*	*Complete*
ACC_3.2-LV4-SS1	$While(State = Active),\ \lvert J_x\rvert \leq J_{Max}$	ACC_3.2	*Passed*	*Complete*
ACC_5.1-LV4-SS1	$While(State = Active),\ 100 \leq Act_\mathrm{Req} \leq 255$	ACC_5.1	*Passed*	*Complete*
SM_3.1-LV4-SS1	$While(State = Active),\ \lvert Act_Req(t) - Act_Req(t-1)\rvert \leq Act_Rate_{Max}$	SM_3.1 (and other similar)	*Passed*	*Complete*
ACC_2.1-LV4-SS2	$While\ ((D_L > D_{Max}\ or\ V_S > V_L)\ and\ \lvert V_S(t) - V_S(t-1)\rvert \leq SS_{Max}),\ V_{Err} \leq V_{ErrMaxSS}$	ACC_2.1	*Passed*	*Complete*
ACC_2.2-LV4-SS2	$While\ ((D_L > D_{Max}\ or\ V_S > V_L)\ and\ \lvert V_S(t) - V_S(t-1)\rvert > SS_{Max}),$	ACC_2.2	*Passed*	*Complete*

	$V_{Err} \leq V_{ErrMax}$			
ACC_2.3-LV4-SS2	$While\ (D_L \leq D_{Max}\ and\ V_S \leq V_L\ and\ \lvert V_S(t) - V_S(t-1)\rvert \leq SS_{Max}),$ $V_{Err} \leq V_{ErrMaxSS}$	ACC_2.3	*Passed*	*Complete*
ACC_2.4-LV4-SS2	$While\ (D_L \leq D_{Max}\ and\ V_S \leq V_L\ and\ \lvert V_S(t) - V_S(t-1)\rvert > SS_{Max}),$ $V_{Err} \leq V_{ErrMax}$	ACC_2.4	*Passed*	*Complete*
ACC_3.1-LV4-SS2	$While(State = Active),$ $\lvert a_x\rvert \leq a_{Max}$	ACC_3.1	*Passed*	*Complete*
ACC_3.2-LV4-SS2	$While(State = Active),$ $\lvert J_x\rvert \leq J_{Max}$	ACC_3.2	*Passed*	*Complete*
ACC_5.1-LV4-SS2	$While(State = Active),$ $100 \leq Act_\mathrm{Req} \leq 255$	ACC_5.1	*Passed*	*Complete*
SM_3.1-LV4-SS2	$While(State = Active),$ $\lvert Act_Req(t) - Act_Req(t-1)\rvert \leq Act_Rate_{Max}$	SM_3.1 (and other similar)	*Passed*	*Complete*
ACC_2.1-LV4-SS3	$While\ ((D_L > D_{Max}\ or\ V_S > V_L)\ and\ \lvert V_S(t) - V_S(t-1)\rvert \leq SS_{Max}),$ $V_{Err} \leq V_{ErrMaxSS}$	ACC_2.1	*Passed*	*Complete*
ACC_2.2-LV4-SS3	$While\ ((D_L > D_{Max}\ or\ V_S > V_L)\ and\ \lvert V_S(t) - V_S(t-1)\rvert > SS_{Max}),$ $V_{Err} \leq V_{ErrMax}$	ACC_2.2	*Passed*	*Complete*
ACC_2.3-LV4-SS3	$While\ (D_L \leq D_{Max}\ and\ V_S \leq V_L\ and\ \lvert V_S(t) - V_S(t-1)\rvert \leq SS_{Max}),$ $V_{Err} \leq V_{ErrMaxSS}$	ACC_2.3	*Passed*	*Complete*
ACC_2.4-LV4-SS3	$While\ (D_L \leq D_{Max}\ and\ V_S \leq V_L\ and\ \lvert V_S(t) - V_S(t-1)\rvert > SS_{Max}),$ $V_{Err} \leq V_{ErrMax}$	ACC_2.4	*Passed*	*Complete*
ACC_3.1-LV4-SS3	$While(State = Active),$ $\lvert a_x\rvert \leq a_{Max}$	ACC_3.1	*Passed*	*Complete*
ACC_3.2-LV4-SS3	$While(State = Active),$	ACC_3.2	*Passed*	*Complete*

	$\|J_x\| \leq J_{Max}$			
ACC_5.1-LV4-SS3	$While(State = Active)$, $100 \leq Act_\text{Req} \leq 255$	ACC_5.1	*Passed*	*Complete*
SM_3.1-LV4-SS3	$While(State = Active)$, $\|Act_Req(t) - Act_Req(t-1)\| \leq Act_Rate_{Max}$	SM_3.1 (and other similar)	*Passed*	*Complete*
ACC_2.1-LV4-SS4	$While\ ((D_L > D_{Max}\ or\ V_S > V_L)\ and\ \|V_S(t) - V_S(t-1)\| \leq SS_{Max})$, $V_{Err} \leq V_{ErrMaxSS}$	ACC_2.1	*Passed*	*Complete*
ACC_2.2-LV4-SS4	$While\ ((D_L > D_{Max}\ or\ V_S > V_L)\ and\ \|V_S(t) - V_S(t-1)\| > SS_{Max})$, $V_{Err} \leq V_{ErrMax}$	ACC_2.2	*Passed*	*Complete*
ACC_2.3-LV4-SS4	$While\ (D_L \leq D_{Max}\ and\ V_S \leq V_L\ and\ \|V_S(t) - V_S(t-1)\| \leq SS_{Max})$, $V_{Err} \leq V_{ErrMaxSS}$	ACC_2.3	*Passed*	*Complete*
ACC_2.4-LV4-SS4	$While\ (D_L \leq D_{Max}\ and\ V_S \leq V_L\ and\ \|V_S(t) - V_S(t-1)\| > SS_{Max})$, $V_{Err} \leq V_{ErrMax}$	ACC_2.4	*Passed*	*Complete*
ACC_3.1-LV4-SS4	$While(State = Active)$, $\|a_x\| \leq a_{Max}$	ACC_3.1	*Passed*	*Complete*
ACC_3.2-LV4-SS4	$While(State = Active)$, $\|J_x\| \leq J_{Max}$	ACC_3.2	*Passed*	*Complete*
ACC_5.1-LV4-SS4	$While(State = Active)$, $100 \leq Act_\text{Req} \leq 255$	ACC_5.1	*Passed*	*Complete*
SM_3.1-LV4-SS4	$While(State = Active)$, $\|Act_Req(t) - Act_Req(t-1)\| \leq Act_Rate_{Max}$	SM_3.1 (and other similar)	*Passed*	*Complete*
ACC_2.1-LV4-SS1	$While\ ((D_L > D_{Max}\ or\ V_S > V_L)\ and\ \|V_S(t) - V_S(t-1)\| \leq SS_{Max})$, $V_{Err} \leq V_{ErrMaxSS}$	ACC_2.1	*Passed*	*Complete*

ACC_2.2-LV4-SS1	$While\ ((D_L > D_{Max}\ or\ V_S > V_L)\ and\ \lvert V_S(t) - V_S(t-1)\rvert > SS_{Max}),\ V_{Err} \leq V_{ErrMax}$	ACC_2.2	*Passed*	*Complete*
ACC_2.3-LV4-SS1	$While\ (D_L \leq D_{Max}\ and\ V_S \leq V_L\ and\ \lvert V_S(t) - V_S(t-1)\rvert \leq SS_{Max}),\ V_{Err} \leq V_{ErrMaxSS}$	ACC_2.3	*Passed*	*Complete*
ACC_2.4-LV4-SS1	$While\ (D_L \leq D_{Max}\ and\ V_S \leq V_L\ and\ \lvert V_S(t) - V_S(t-1)\rvert > SS_{Max}),\ V_{Err} \leq V_{ErrMax}$	ACC_2.4	*Passed*	*Complete*
ACC_3.1-LV4-SS5	$While(State = Active),\ \lvert a_x\rvert \leq a_{Max}$	ACC_3.1	*Passed*	*Complete*
ACC_3.2-LV4-SS5	$While(State = Active),\ \lvert J_x\rvert \leq J_{Max}$	ACC_3.2	*Passed*	*Complete*
ACC_5.1-LV4-SS5	$While(State = Active),\ 100 \leq Act_\mathrm{Req} \leq 255$	ACC_5.1	*Passed*	*Complete*
SM_3.1-LV4-SS5	$While(State = Active),\ \lvert Act_Req(t) - Act_Req(t-1)\rvert \leq Act_Rate_{Max}$	SM_3.1 (and other similar)	*Passed*	*Complete*

LV5 – Erratic Vehicle

Table 9. LV5 Erractic Vehicle Traceability Matrix.

Test ID	**Expected Outputs**	**Associated Requirement(s)**	**Result**	**Test Status**
ACC_2.1-LV5-SS1	$While\ ((D_L > D_{Max}\ or\ V_S > V_L)\ and\ \lvert V_S(t) - V_S(t-1)\rvert \leq SS_{Max}),\ V_{Err} \leq V_{ErrMaxSS}$	ACC_2.1	*Passed*	*Complete*
ACC_2.2-LV5-SS1	$While\ ((D_L > D_{Max}\ or\ V_S > V_L)\ and\ \lvert V_S(t) - V_S(t-1)\rvert > SS_{Max}),\ V_{Err} \leq V_{ErrMax}$	ACC_2.2	*Passed*	*Complete*

ACC_2.3-LV5-SS1	$While\ (D_L \leq D_{Max}\ and\ V_S \leq V_L\ and\ \lvert V_S(t) - V_S(t-1)\rvert \leq SS_{Max}),\ V_{Err} \leq V_{ErrMaxSS}$	ACC_2.3	*Passed*	*Complete*
ACC_2.4-LV5-SS1	$While\ (D_L \leq D_{Max}\ and\ V_S \leq V_L\ and\ \lvert V_S(t) - V_S(t-1)\rvert > SS_{Max}),\ V_{Err} \leq V_{ErrMax}$	ACC_2.4	*Passed*	*Complete*
ACC_3.1-LV5-SS1	$While(State = Active),\ \lvert a_x\rvert \leq a_{Max}$	ACC_3.1	*Passed*	*Complete*
ACC_3.2-LV5-SS1	$While(State = Active),\ \lvert J_x\rvert \leq J_{Max}$	ACC_3.2	*Passed*	*Complete*
ACC_5.1-LV5-SS1	$While(State = Active),\ 100 \leq Act_\mathrm{Req} \leq 255$	ACC_5.1	*Passed*	*Complete*
SM_3.1-LV5-SS1	$While(State = Active),\ \lvert Act_Req(t) - Act_Req(t-1)\rvert \leq Act_Rate_{Max}$	SM_3.1 (and other similar)	*Passed*	*Complete*
ACC_2.1-LV5-SS2	$While\ ((D_L > D_{Max}\ or\ V_S > V_L)\ and\ \lvert V_S(t) - V_S(t-1)\rvert \leq SS_{Max}),\ V_{Err} \leq V_{ErrMaxSS}$	ACC_2.1	*Passed*	*Complete*
ACC_2.2-LV5-SS2	$While\ ((D_L > D_{Max}\ or\ V_S > V_L)\ and\ \lvert V_S(t) - V_S(t-1)\rvert > SS_{Max}),\ V_{Err} \leq V_{ErrMax}$	ACC_2.2	*Passed*	*Complete*
ACC_2.3-LV5-SS2	$While\ (D_L \leq D_{Max}\ and\ V_S \leq V_L\ and\ \lvert V_S(t) - V_S(t-1)\rvert \leq SS_{Max}),\ V_{Err} \leq V_{ErrMaxSS}$	ACC_2.3	*Passed*	*Complete*
ACC_2.4-LV5-SS2	$While\ (D_L \leq D_{Max}\ and\ V_S \leq V_L\ and\ \lvert V_S(t) - V_S(t-1)\rvert > SS_{Max}),\ V_{Err} \leq V_{ErrMax}$	ACC_2.4	*Passed*	*Complete*
ACC_3.1-LV5-SS2	$While(State = Active),\ \lvert a_x\rvert \leq a_{Max}$	ACC_3.1	*Passed*	*Complete*
ACC_3.2-LV5-SS2	$While(State = Active),$	ACC_3.2	*Passed*	*Complete*

	$\|J_x\| \leq J_{Max}$			
ACC_5.1-LV5-SS2	$While(State = Active)$, $100 \leq Act_\text{Req} \leq 255$	ACC_5.1	*Passed*	*Complete*
SM_3.1-LV5-SS2	$While(State = Active)$, $\|Act_Req(t) - Act_Req(t-1)\| \leq Act_Rate_{Max}$	SM_3.1 (and other similar)	*Passed*	*Complete*
ACC_2.1-LV5-SS3	$While\ ((D_L > D_{Max}\ or\ V_S > V_L)\ and\ \|V_S(t) - V_S(t-1)\| \leq SS_{Max})$, $V_{Err} \leq V_{ErrMaxSS}$	ACC_2.1	*Passed*	*Complete*
ACC_2.2-LV5-SS3	$While\ ((D_L > D_{Max}\ or\ V_S > V_L)\ and\ \|V_S(t) - V_S(t-1)\| > SS_{Max})$, $V_{Err} \leq V_{ErrMax}$	ACC_2.2	*Passed*	*Complete*
ACC_2.3-LV5-SS3	$While\ (D_L \leq D_{Max}\ and\ V_S \leq V_L\ and\ \|V_S(t) - V_S(t-1)\| \leq SS_{Max})$, $V_{Err} \leq V_{ErrMaxSS}$	ACC_2.3	*Passed*	*Complete*
ACC_2.4-LV5-SS3	$While\ (D_L \leq D_{Max}\ and\ V_S \leq V_L\ and\ \|V_S(t) - V_S(t-1)\| > SS_{Max})$, $V_{Err} \leq V_{ErrMax}$	ACC_2.4	*Passed*	*Complete*
ACC_3.1-LV5-SS3	$While(State = Active)$, $\|a_x\| \leq a_{Max}$	ACC_3.1	*Passed*	*Complete*
ACC_3.2-LV5-SS3	$While(State = Active)$, $\|J_x\| \leq J_{Max}$	ACC_3.2	*Passed*	*Complete*
ACC_5.1-LV5-SS3	$While(State = Active)$, $100 \leq Act_\text{Req} \leq 255$	ACC_5.1	*Passed*	*Complete*
SM_3.1-LV5-SS3	$While(State = Active)$, $\|Act_Req(t) - Act_Req(t-1)\| \leq Act_Rate_{Max}$	SM_3.1 (and other similar)	*Passed*	*Complete*
ACC_2.1-LV5-SS4	$While\ ((D_L > D_{Max}\ or\ V_S > V_L)\ and\ \|V_S(t) - V_S(t-1)\| \leq SS_{Max})$, $V_{Err} \leq V_{ErrMaxSS}$	ACC_2.1	*Passed*	*Complete*

ACC_2.2-LV5-SS4	$While\ ((D_L > D_{Max}\ or\ V_S > V_L)\ and\ \lvert V_S(t) - V_S(t-1)\rvert > SS_{Max}),\ V_{Err} \leq V_{ErrMax}$	ACC_2.2	*Passed*	*Complete*
ACC_2.3-LV5-SS4	$While\ (D_L \leq D_{Max}\ and\ V_S \leq V_L\ and\ \lvert V_S(t) - V_S(t-1)\rvert \leq SS_{Max}),\ V_{Err} \leq V_{ErrMaxSS}$	ACC_2.3	*Passed*	*Complete*
ACC_2.4-LV5-SS4	$While\ (D_L \leq D_{Max}\ and\ V_S \leq V_L\ and\ \lvert V_S(t) - V_S(t-1)\rvert > SS_{Max}),\ V_{Err} \leq V_{ErrMax}$	ACC_2.4	*Passed*	*Complete*
ACC_3.1-LV5-SS4	$While(State = Active),\ \lvert a_x\rvert \leq a_{Max}$	ACC_3.1	*Passed*	*Complete*
ACC_3.2-LV5-SS4	$While(State = Active),\ \lvert J_x\rvert \leq J_{Max}$	ACC_3.2	*Passed*	*Complete*
ACC_5.1-LV5-SS4	$While(State = Active),\ 100 \leq Act_\mathrm{Req} \leq 255$	ACC_5.1	*Passed*	*Complete*
SM_3.1-LV5-SS4	$While(State = Active),\ \lvert Act_Req(t) - Act_Req(t-1)\rvert \leq Act_Rate_{Max}$	SM_3.1 (and other similar)	*Passed*	*Complete*
ACC_2.1-LV5-SS1	$While\ ((D_L > D_{Max}\ or\ V_S > V_L)\ and\ \lvert V_S(t) - V_S(t-1)\rvert \leq SS_{Max}),\ V_{Err} \leq V_{ErrMaxSS}$	ACC_2.1	*Passed*	*Complete*
ACC_2.2-LV5-SS1	$While\ ((D_L > D_{Max}\ or\ V_S > V_L)\ and\ \lvert V_S(t) - V_S(t-1)\rvert > SS_{Max}),\ V_{Err} \leq V_{ErrMax}$	ACC_2.2	*Passed*	*Complete*
ACC_2.3-LV5-SS1	$While\ (D_L \leq D_{Max}\ and\ V_S \leq V_L\ and\ \lvert V_S(t) - V_S(t-1)\rvert \leq SS_{Max}),\ V_{Err} \leq V_{ErrMaxSS}$	ACC_2.3	*Passed*	*Complete*
ACC_2.4-LV5-SS1	$While\ (D_L \leq D_{Max}\ and\ V_S \leq V_L\ and\ \lvert V_S(t) - V_S(t-1)\rvert > SS_{Max}),\ V_{Err} \leq V_{ErrMax}$	ACC_2.4	*Passed*	*Complete*
ACC_3.1-LV5-SS5	$While(State = Active),$	ACC_3.1	*Passed*	*Complete*

	$\lvert a_x \rvert \leq a_{Max}$			
ACC_3.2-LV5-SS5	$While(State = Active)$, $\lvert J_x \rvert \leq J_{Max}$	ACC_3.2	*Passed*	*Complete*
ACC_5.1-LV5-SS5	$While(State = Active)$, $100 \leq Act_\text{Req} \leq 255$	ACC_5.1	*Passed*	*Complete*
SM_3.1-LV5-SS5	$While(State = Active)$, $\lvert Act_Req(t) - Act_Req(t-1) \rvert \leq Act_Rate_{Max}$	SM_3.1 (and other similar)	*Passed*	*Complete*

Additional System Tests

The following tests will not use test sequences but instead will use their own unique set of inputs.

Table 10. Additional System Tests Traceability Matrix.

Test ID	Inputs	Expected Outputs	Associated Requirement(s)	Result	Test Status
ACC_4.1-1	At steady state while both lead and ego vehicle are travelling at 0.2 cm/s, the ego vehicle suddenly brakes with maximum deceleration until stopped.	Ego vehicle stops ($V_E = 0$) without collision ($D_L > 0$)	ACC_4.1	*Passed*	*Complete*
ACC_4.1-2	At steady state while both lead and ego vehicle are travelling at 0.5 cm/s, the ego vehicle suddenly brakes with maximum deceleration until stopped.	Ego vehicle stops ($V_E = 0$) without collision ($D_L > 0$)	ACC_4.1	*Passed*	*Complete*
ACC_4.1-3	At steady state while both lead and ego vehicle are travelling at 0.8 cm/s, the ego vehicle	Ego vehicle stops ($V_E = 0$)	ACC_4.1	*Passed*	*Complete*

	suddenly brakes with maximum deceleration until stopped.	without collision ($D_L > 0$)			

Unit Tests

Table 11. Safety Monitor Traceability Matrix.

Test ID	Requirement	Test Description	Test Method	Pass Criteria	Result	Test Status
SM_4-3	SM_4	Verify the system checks are implemented correctly	Check the implemented values of the CRC standard used with Expected Values for a given set of inputs. Similarly for the RC	CRC processed = Expected Value	Passed	*Complete*
SM_4-4				RC processed = RC expected	Passed	*Complete*

Simulator

Table 12. Simulator Traceability Matrix.

Test ID	Requirement	Test Description	Test Method	Pass Criteria	Result	Test Status
Sim_1.2-1	Sim_1.2	Verify the generation of each of the test sequences for lead velocity	Use the simulator to generate a lead velocity profile that matches the description of the test sequence	Generates LV-1	*Passed*	*Complete*
Sim_1.2-2				Generates LV-2	*Passed*	*Complete*
Sim_1.2-3				Generates LV-3	*Passed*	*Complete*
Sim_1.2-4				Generates LV-4	*Passed*	*Complete*
Sim_1.2-5				Generates LV-5	*Passed*	*Complete*
Sim_1.1-1	Sim_1.1	Verify that the lead distance is	Run each of the test sequences from the system tests and graph lead velocity vs time and lead	The calculated lead distance from the simulator matches the	*Passed*	*Complete*

		correctly calculated	distance over time and compare with manually pre-computed values (assume ego velocity is zero)	manual calculations (within acceptable error)		
Sim_1.3-1	Sim_1.3	Verify the generation of each of the test sequences for set speed	Use the simulator to generate a set speed profile that matches the description of the test sequence	Generates SS-1	*Passed*	*Complete*
Sim_1.3-2				Generates SS-2	*Passed*	*Complete*
Sim_1.3-3				Generates SS-3	*Passed*	*Complete*
Sim_1.3-4				Generates SS-4	*Passed*	*Complete*
Sim_1.3-5				Generates SS-5	*Passed*	*Complete*

Hardware Verification

Sim Dyno

Table 13. SimDyno Hardware Traceability Matrix.

Test ID	**Requirement**	**Test Description**	**Test Method**	**Pass Criteria**	**Result**	**Test Status**
Dyno_1.1	Dyno_1	Verify that the dynamometer correctly measures and reports wheel & treadmill speeds.	Compare dynamometer hall-effect sensor readings with on-board vehicle hall effect sensor.	Dynamometer hall effect sensor readings match on-board vehicle hall effect sensors within +- 5% error.	The dynamometer hall effect sensors matched the on-board hall effect sensors within the limit.	*Complete*
Dyno_2.1	Dyno_2 (Lateral Constraint)	Ensure RC cars do not move laterally when placed on the dynamometer.	Apply lateral force and observe movement.	Limited lateral displacement beyond +- 5 mm.	The RC vehicle does not move laterally when	Complete

					placed on the dynamometer.	
Dyno_3.1	Dyno_3 (Vibration Dampening)	Verify dynamometer reduces vibrations at the test bench.	Observe if the dynamometer is resilient to vibrations when applying shaking motion to the dynamometer.	Qualitative observation of the vibrations of the dynamometer. Vibrations of dynamometer would not affect normal vehicle operation.	The dynamometer is rigid enough that small vibrations do not disturb vehicle operation. A chair seat-pad was also added underneath to further reduce this.	*Complete*
Dyno_4.1	Dyno_4 (Low-friction Setup)	Ensure the system minimizes friction losses.	Measure the difference between wheel and treadmill speeds.	W1 – T1 < Dyn(max) and W2 – T2 < Dyn(max).	The difference between the wheel and treadmill speeds was less than the pre-determined limit.	*Complete*
Dyno_5.1	Dyno_5 (Secure Mounting)	Verify that the safety monitor is securely mounted.	Fasten the safety monitor to its mounting location and apply external pressure.	No misalignment or detachment after testing.	*Passed*	*Complete*

Dyno_5.2		Verify that the hall effect sensors are securely mounted.	Fasten the hall effect sensors to their mounting location and apply external pressure.	No misalignment or detachment after testing.	*Passed*	*Complete*
Dyno_5.3		Verify that the magnets are securely mounted.	Fasten the magnets to their mounting location and apply external pressure.	No misalignment or detachment after testing.	*Passed*	*Complete*
Dyno_5.4		Verify that the wired connections of the safety monitor are securely connected.	Apply external pressure to the wires of the safety monitor.	No misalignment or detachment after testing.	*Passed*	*Complete*
Dyno_5.5		Verify that the wired connections of the RC vehicle are securely connected.	Apply external pressure to the wires of the RC vehicle.	No misalignment or detachment after testing.	*Passed*	*Complete*
ACC_6.1	ACC_6 (and all children)	Verify that the ego speed is calculated properly from the wheel speed sensors.	Generate values for each of the four speed sensors such that all 11 possible combinations of agreement are compared (5 sets of 2, 4 sets of 3, 1 set of 1)	The final calculated velocity is according to the criteria in ACC_6.		*Incomplete*

Safety Monitor

Table 14. Safety Monitor Hardware Traceability Matrix.

Test ID	Requirement	Test Description	Test Method	Pass Criteria	Result	Test Status
SM_5.1.1	SM_5	The safety monitor shall be protected from external forces acting upon connectors.	External pressure shall be applied to the case.	The case protects the safety monitors from external forces.	*Passed*	*Complete*
SM_5.1.1		The safety monitor case shall cover the printed circuit board to prevent any possible accidental shorting.	The safety monitor case shall be analyzed for possible	The case protects the printed circuit board to a reasonable extent (necessary connectors must still be accessible).	*Passed*	*Complete*
SM_6.1.1	SM_6	The safety monitor shall be mounted close to the microcontroller of the vehicle to simplify wiring and troubleshooting.	The monitor shall be mounted in various feasible regions that minimize complicated wiring harnesses.	The monitor can be mounted in a region that facilitates easy wiring and troubleshooting.	*Passed*	*Complete*
SM_7.1.1	SM_7	The ethernet bus shall provide proper	An ethernet bus shall be connected between the simulator computer	The serial monitor correctly receives the message	*Passed*	*Complete* (Tested with “123” and

		communication between the computer and the safety monitor.	and the safety monitor with TCP requests sent between the two nodes.	transferred from the computer and vice versa.		received on both devices appropriately)

Hall Effect Sensors

Table 15. Hall Effect Sensors Hardware Traceability Matrix.

Test ID	Requirement	Test Description	Test Method	Pass Criteria	Result	Test Status
HE_1.1.1	HE_1.1	The duty cycle measurements shall be properly communicated to the on-board Arduino.	An Arduino script shall be initialized to print the duty cycle input from the serial monitor.	If the duty cycle measurements from the serial monitor vary from 50% (the nominal duty cycle of the hall effect sensor) it is behaving correctly.	*Passed*	*Complete*
HE_2.1.1	HE_2.1	The hall effect sensors must be mounted close enough to the rotating magnets.	The hall effect sensors shall be mounted at various distances and duty cycle measurements will be measured through the Arduino serial monitor.	The duty cycle measurements align with the rotations of the magnets and are distinct enough.	*Passed*	*Complete*

Remote Control Vehicle

Table 16. Remote Control Vehicle Hardware Traceability Matrix.

Test ID	Requirement	Test Description	Test Method	Pass Criteria	Result	Test Status
RC_1.1.1	RC_1.1	The wheels of the RC vehicle shall rotate upon command from the on-board Arduino.	A PWM signal shall be transferred from the safety monitor to the RC vehicle over UART that shall rotate the wheels of the vehicle.	The wheels of the vehicle rotate based on the PWM value.	*Passed*	*Complete*
RC_2.1.1	RC_2.1	The UART bus shall provide proper communication between the safety monitor and the on-board Arduino.			*Passed*	*Complete*

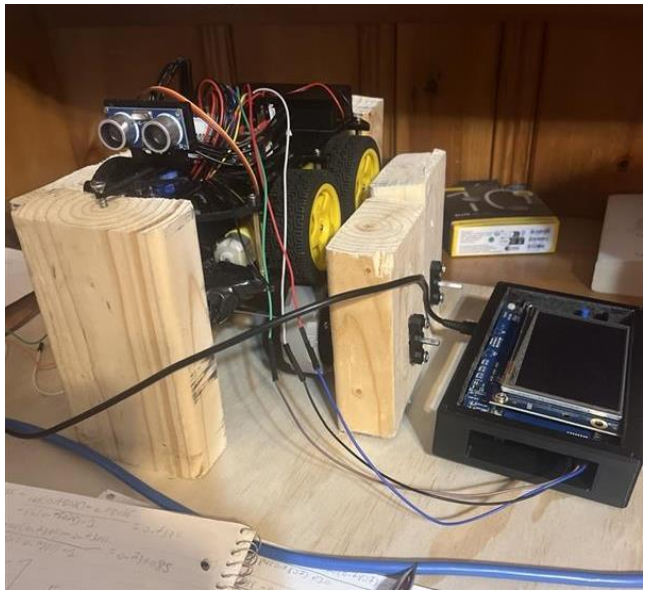


Figure 2. Assembly of Wired Dynamometer Platform with Safety Monitor in Case.

Test Coverage

The test sequence for system tests was designed to maximize the amount of coverage of the input domain while considering realistic behaviour of the system. The HWFET and UDDS are both standardized test inputs for drive cycles used in the automotive domain, specifically because they reflect the average highway and city drive respectively and are long enough to take into account a wide range of different scenarios within that. The stepwise velocity sequence was designed to cover the extremes – both in that the minimum and maximum velocities are tested, as well as quite large jumps which both decrease and increase. Finally, the erratic vehicle test sequence, which randomly generates each time, will be conducted as many times as deemed necessary to reasonably conclude the system is effective. We predict this test sequence will point out obscure bugs we would otherwise have missed.

Similarly, for set speed, three test sequences are designed to test typical drives scenarios, with constant average meant to represent a city drive, constant maximum a highway drive, and constant minimum to test the lower boundaries of the system. As before, the stepwise sequence will test large jumps in set speed, and random will be used to fill in any gaps from the other test sequences. The unique interactions between the set speed sequences and lead velocity will introduce many more scenarios, such as when the ACC system changes from cruise to follow mode and back. A possible gap in the test space is the speeds between 0.2, 0.5 and 0.8 cm/s. We believe these three speeds will be sufficient to cover all behaviour of the system but will ensure intermediate speeds work as well.

Some requirements, such as ACC_4.1, are only relevant in a specific set of circumstances, so specific tests were designed to test those. A similar principle of starting from the average, maximum and minimum velocities was used to test these, though gaps may exist in between these which will be considered.

Many of the tests for the safety monitor and sim dyno always apply, so unit tests were designed to ensure proper behaviour during any run cycle, though these requirements will be carefully monitored during all run cycles to ensure they are still working properly.

Appendix A: SIL Test Results

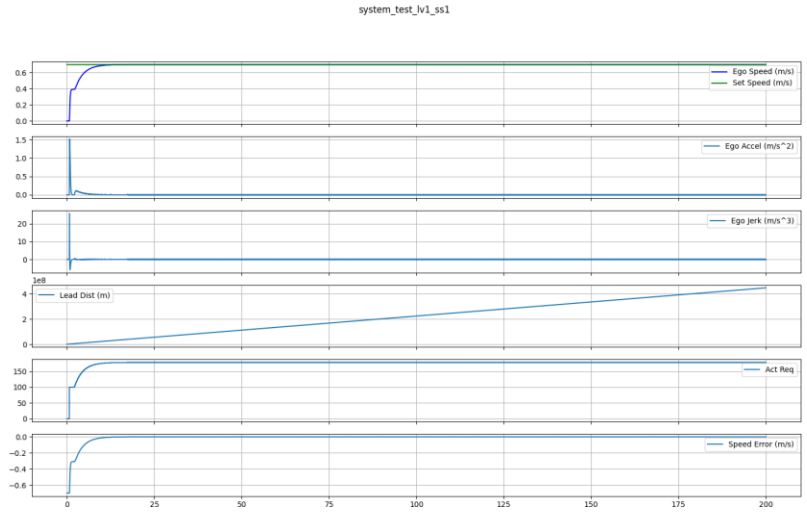


Figure 3

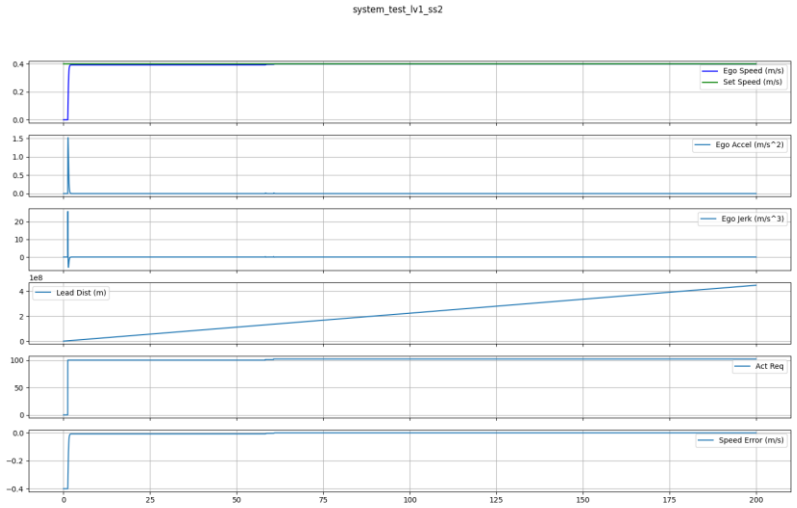


Figure 4

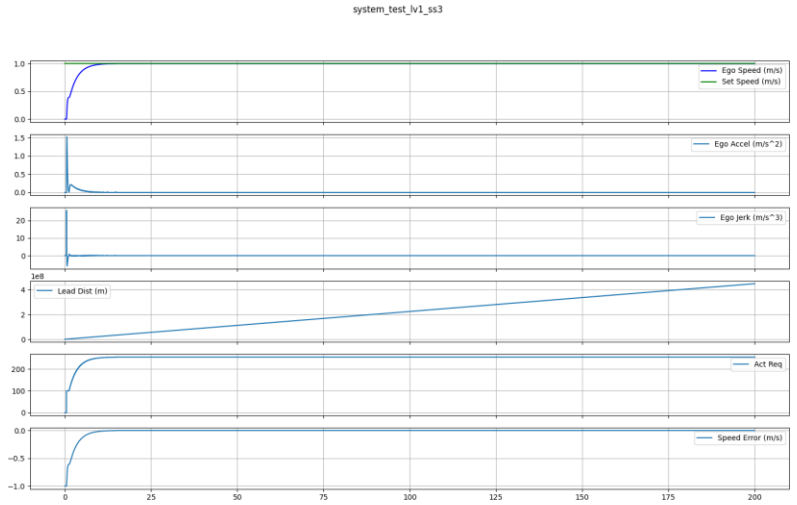


Figure 5

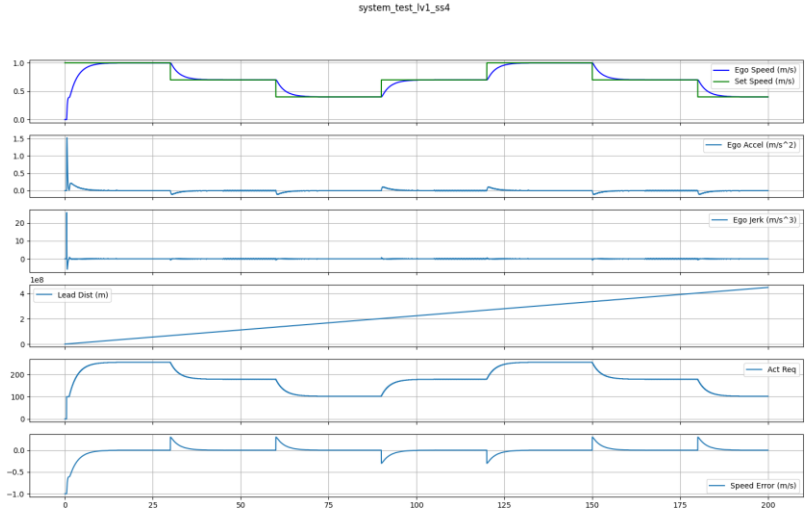


Figure 6

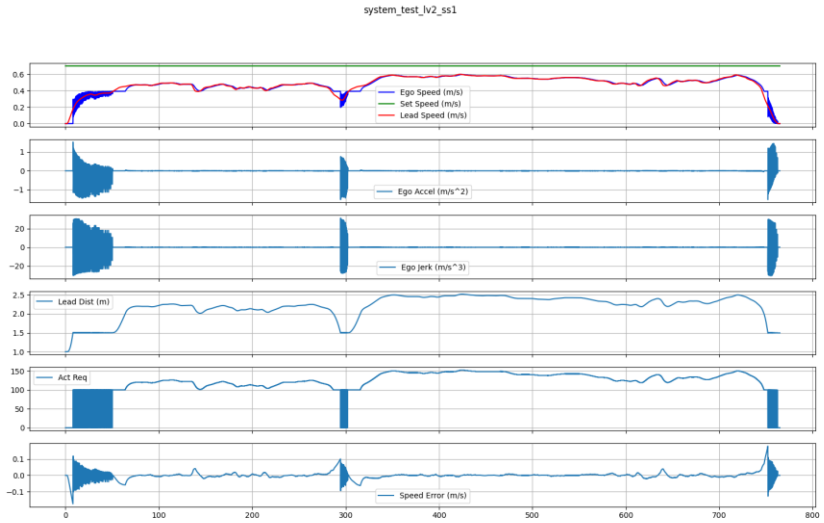


Figure 7

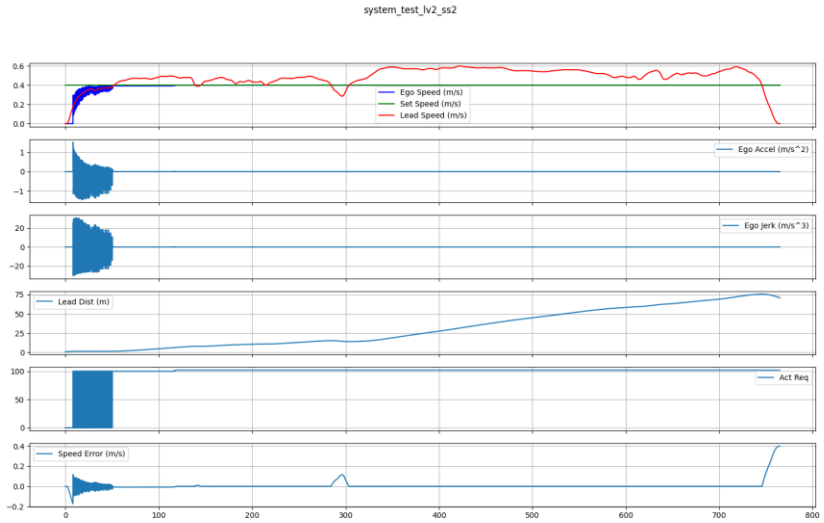

system_test_lv2_ss2
Ego Speed (m/s)
Set Speed (m/s)
Lead Speed (m/s)
Ego Accel (m/s^2)
Ego Jerk (m/s^3)
Lead Dist (m)
Act Req
Speed Error (m/s)

Figure 8

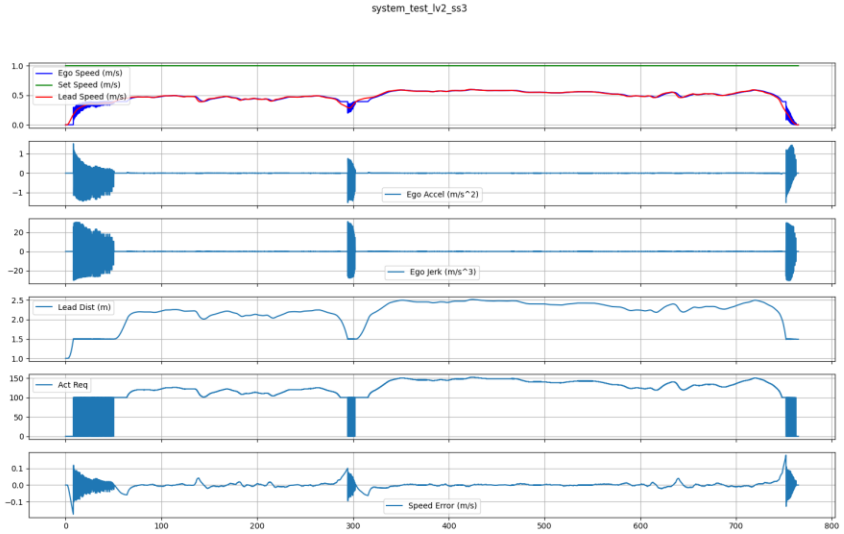

system_test_lv2_ss3
Ego Speed (m/s)
Set Speed (m/s)
Lead Speed (m/s)
Ego Accel (m/s^2)
Ego Jerk (m/s^3)
Lead Dist (m)
Act Req
Speed Error (m/s)

Figure 9

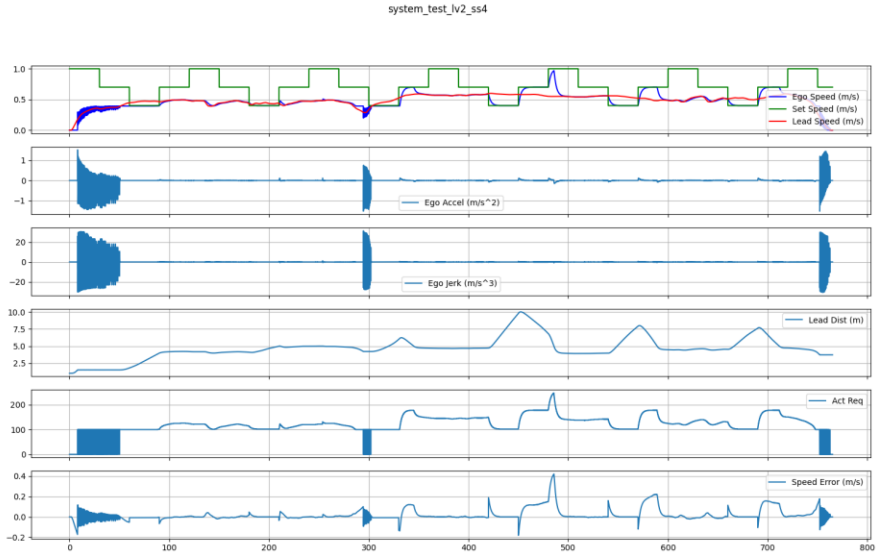

system_test_lv2_ss4
Ego Speed (m/s)
Set Speed (m/s)
Lead Speed (m/s)
Ego Accel (m/s^2)
Ego Jerk (m/s^3)
Lead Dist (m)
Act Req
Speed Error (m/s)

Figure 10

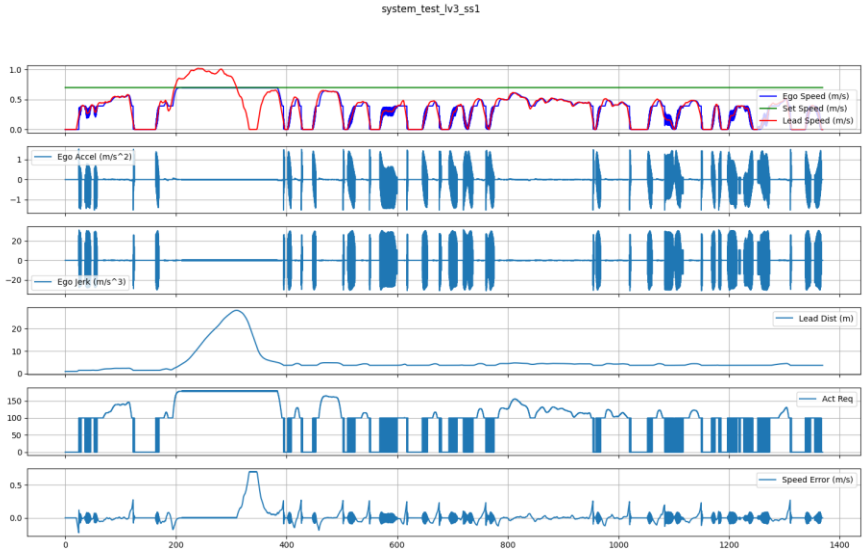

system_test_lv3_ss1
Ego Speed (m/s)
Set Speed (m/s)
Lead Speed (m/s)
Ego Accel (m/s^2)
Ego jerk (m/s^3)
Lead Dist (m)
Act Req
Speed Error (m/s)

Figure 11

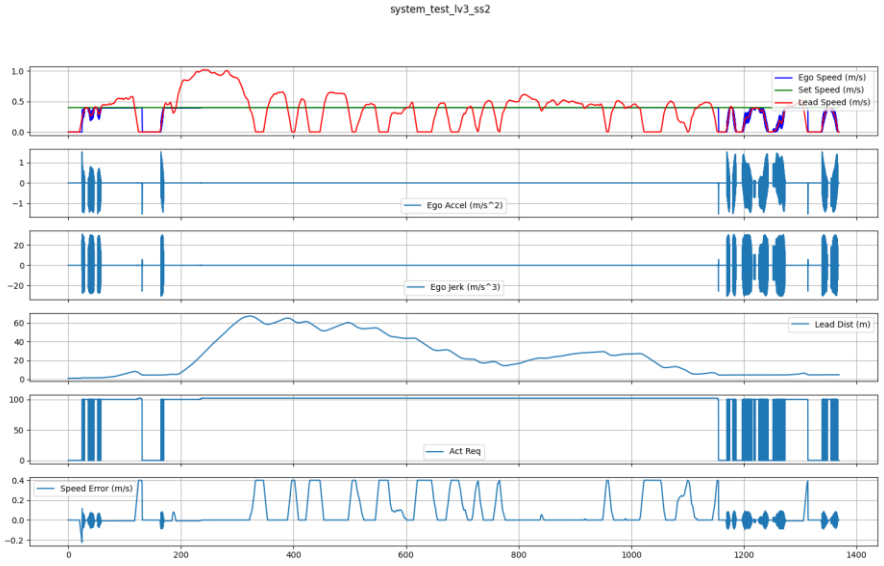

system_test_lv3_ss2
Ego Speed (m/s)
Set Speed (m/s)
Lead Speed (m/s)
Ego Accel (m/s^2)
Ego Jerk (m/s^3)
Lead Dist (m)
Act Req
Speed Error (m/s)

Figure 12

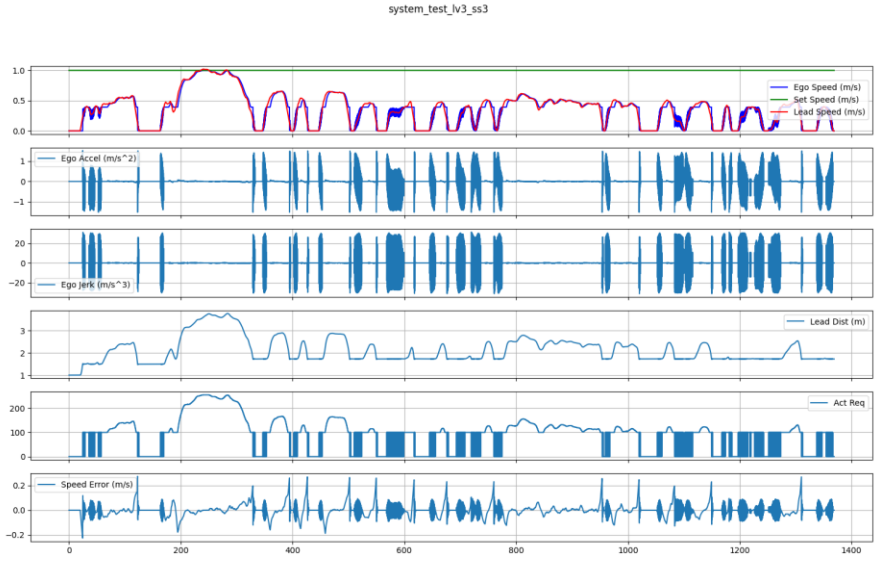

system_test_lv3_ss3
Ego Speed (m/s)
Set Speed (m/s)
Lead Speed (m/s)
Ego Accel (m/s^2)
Ego jerk (m/s^3)
Lead Dist (m)
Act Req
Speed Error (m/s)

Figure 13

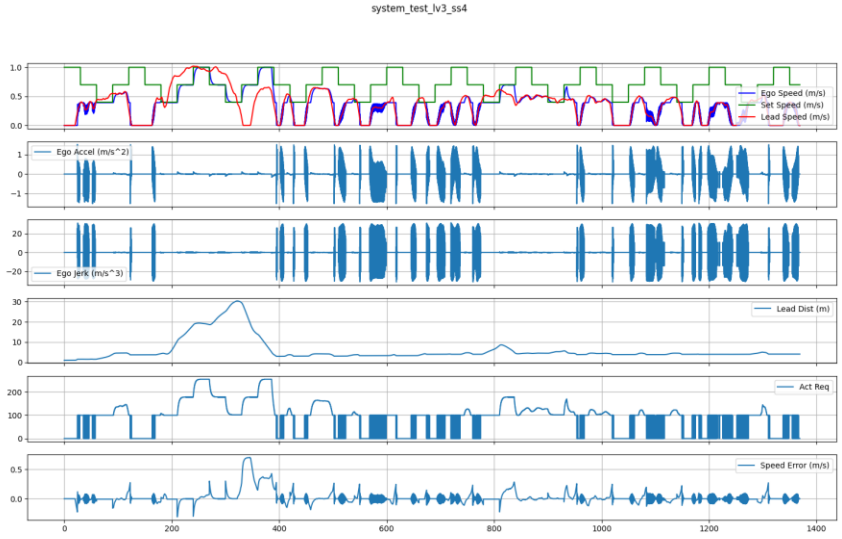

system_test_lv3_ss4
Ego Speed (m/s)
Set Speed (m/s)
Lead Speed (m/s)
Ego Accel (m/s^2)
Ego Jerk (m/s^3)
Lead Dist (m)
Act Req
Speed Error (m/s)

Figure 14

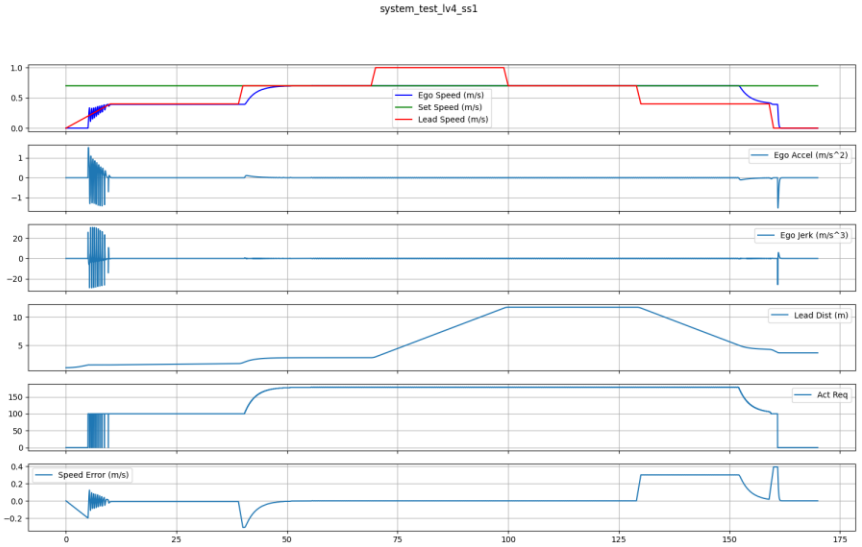

system_test_lv4_ss1
Ego Speed (m/s)
Set Speed (m/s)
Lead Speed (m/s)
Ego Accel (m/s^2)
Ego Jerk (m/s^3)
Lead Dist (m)
Act Req
Speed Error (m/s)

Figure 15

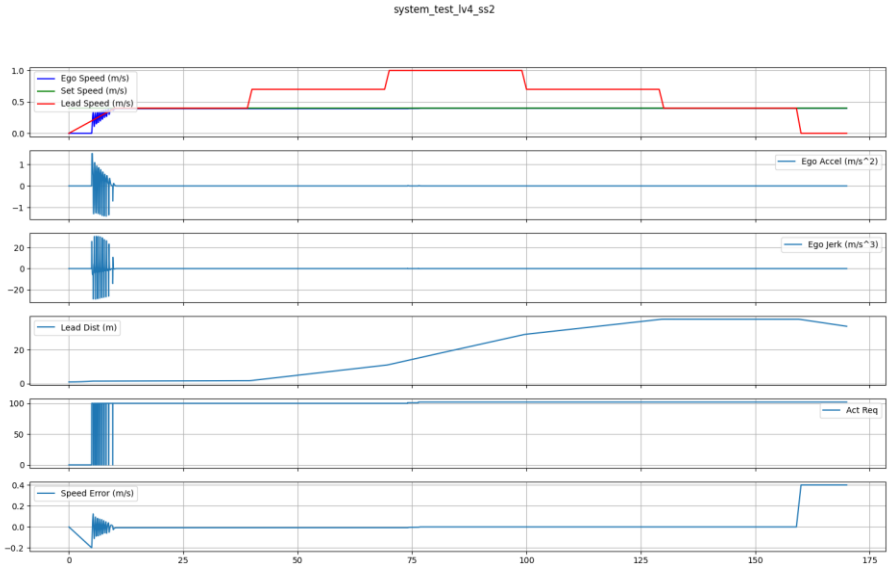

system_test_lv4_ss2
Ego Speed (m/s)
Set Speed (m/s)
Lead Speed (m/s)
Ego Accel (m/s^2)
Ego Jerk (m/s^3)
Lead Dist (m)
Act Req
Speed Error (m/s)

Figure 16

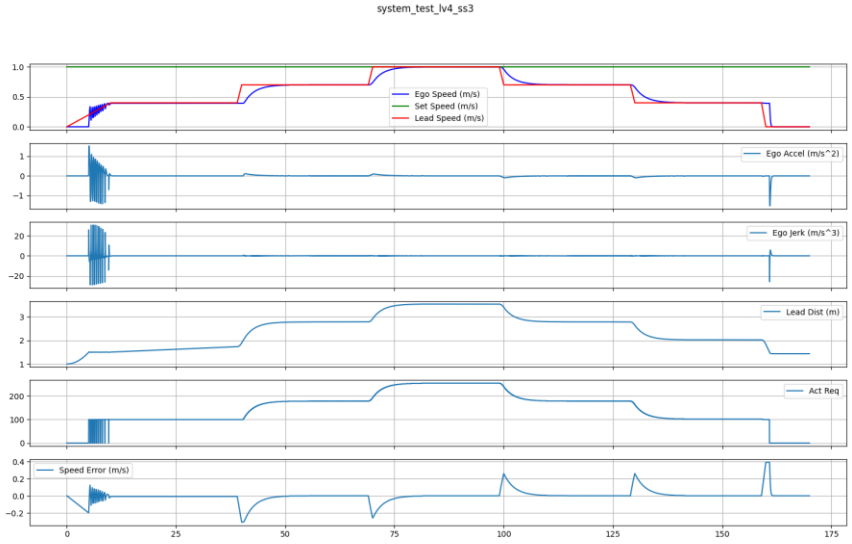

system_test_lv4_ss3
Ego Speed (m/s)
Set Speed (m/s)
Lead Speed (m/s)
Ego Accel (m/s^2)
Ego Jerk (m/s^3)
Lead Dist (m)
Act Req
Speed Error (m/s)

Figure 17

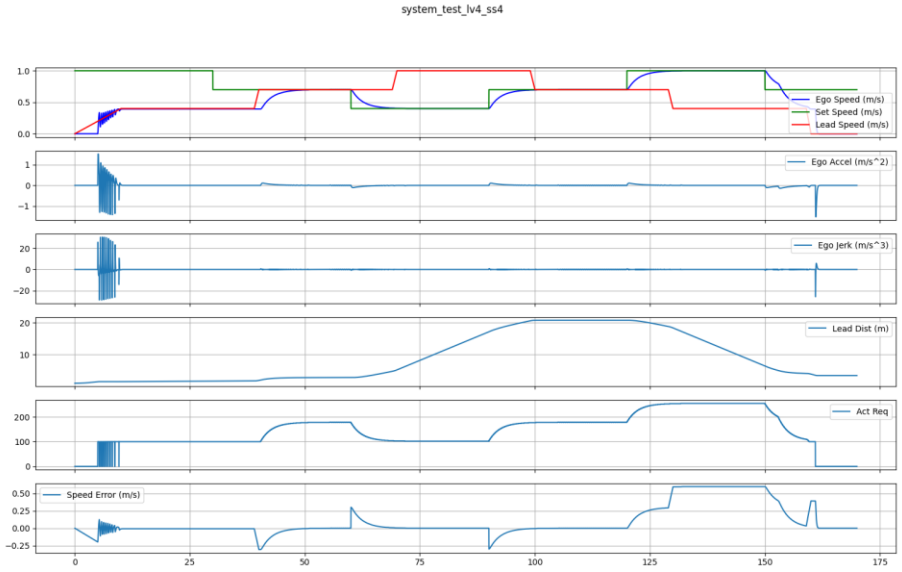

system_test_lv4_ss4
Ego Speed (m/s)
Set Speed (m/s)
Lead Speed (m/s)
Ego Accel (m/s^2)
Ego Jerk (m/s^3)
Lead Dist (m)
Act Req
Speed Error (m/s)

Figure 18

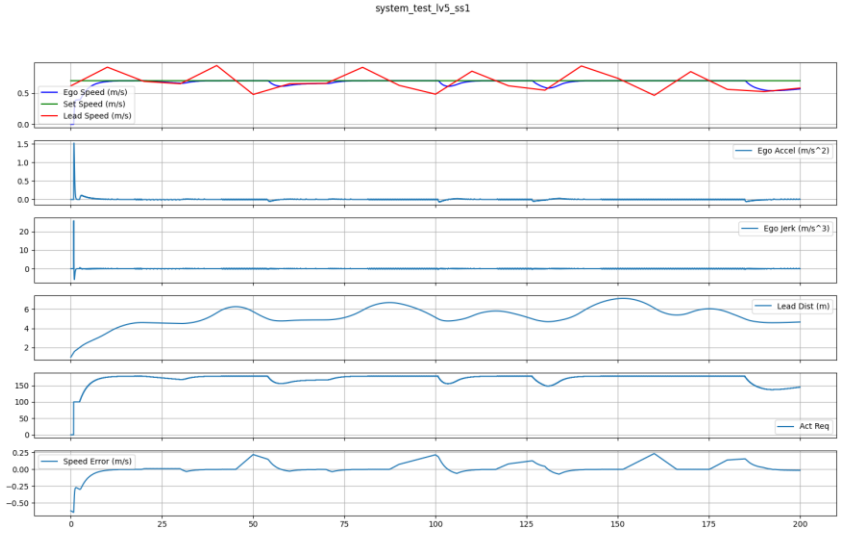

system_test_lv5_ss1
Ego Speed (m/s)
Set Speed (m/s)
Lead Speed (m/s)
Ego Accel (m/s^2)
Ego Jerk (m/s^3)
Lead Dist (m)
Act Req
Speed Error (m/s)

Figure 19

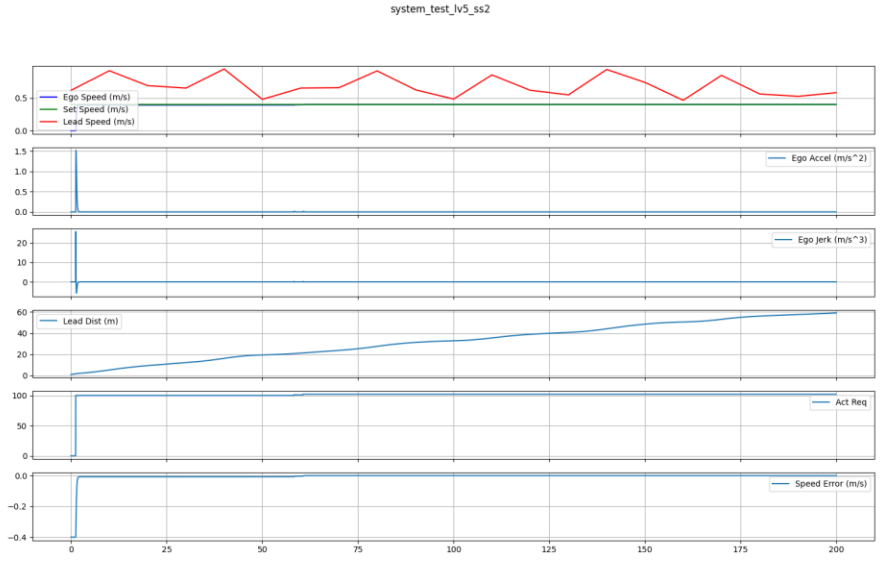

system_test_lv5_ss2
Ego Speed (m/s)
Set Speed (m/s)
Lead Speed (m/s)
Ego Accel (m/s^2)
Ego Jerk (m/s^3)
Lead Dist (m)
Act Req
Speed Error (m/s)

Figure 20

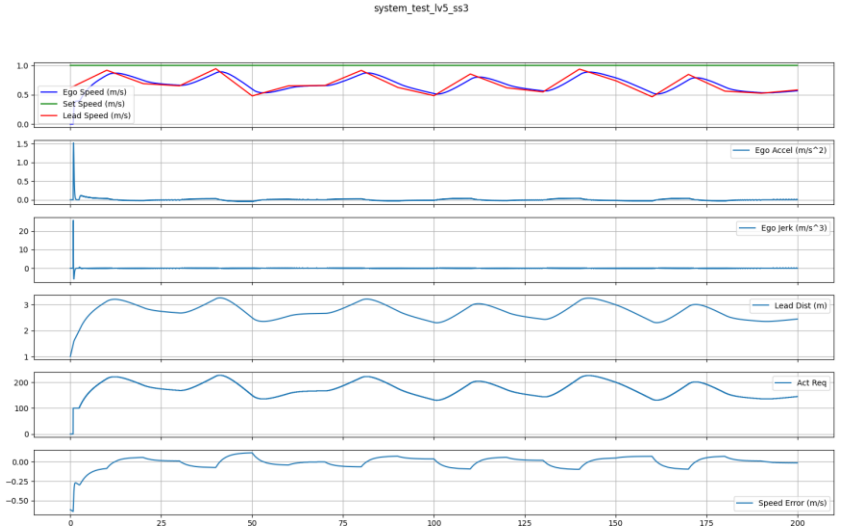

system_test_lv5_ss3
Ego Speed (m/s)
Set Speed (m/s)
Lead Speed (m/s)
Ego Accel (m/s^2)
Ego Jerk (m/s^3)
Lead Dist (m)
Act Req
Speed Error (m/s)

Figure 21

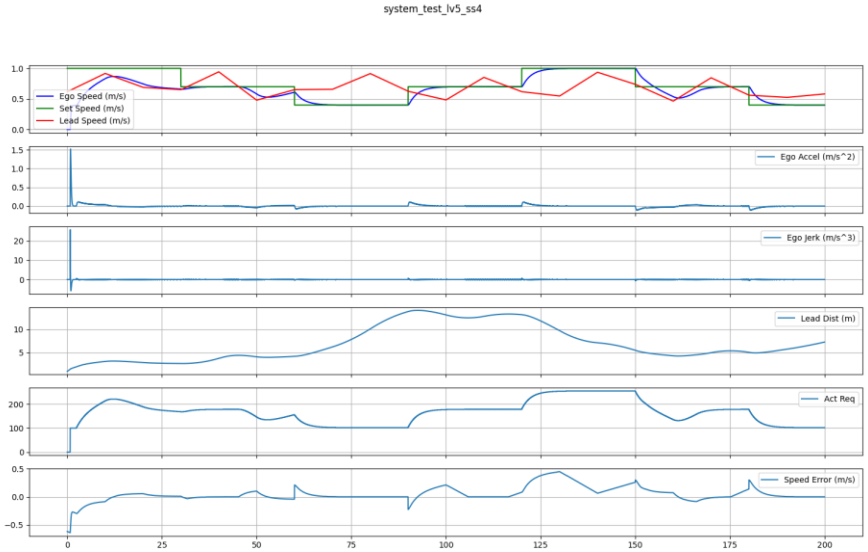

system_test_lv5_ss4
Ego Speed (m/s)
Set Speed (m/s)
Lead Speed (m/s)
Ego Accel (m/s^2)
Ego Jerk (m/s^3)
Lead Dist (m)
Act Req
Speed Error (m/s)

Figure 22